

Text Summariser powered by a fine-tuned LLM, FLAN-T5 Base

Introduction

Natural language processing is one of the most important area in artificial intelligence, enabling computers to understand, process, and generate human-like responses. One of the important application is text summarization, which converts a lengthy textual information to a concise meaningful summary.

This projects specifically demonstrates dialogue summarization, that presents additional challenges due to informal language, multiple speakers and contextual dependencies used in a text messages. These quirks require models to capture semantic relationship between sentences and generate meaningful outputs, rather than copy sentences from the text.

In this project FLAN-T5 Base model is selected because its an instruction tuned model that has high capability of generating high quality summaries from natural language instructions. The model is fine-tuned using the SAMSum dataset that contains messages, chats and dialogues paired with human written summaries, making this dataset the best option for this project.

Methodology

- 1) Dataset:** SAMSum dialogues summarization dataset was used in this project. It contains messenger style conversation with dialogue and a human written summary. The dataset contains train, test and validation splits.
- 2) Preprocessing:** Each dialogue was converted to an instruction based prompt that's compatible with T5's architecture. This format acts as a leverage to T5's instruction tuning, and enable model to do the specific task that I require.
- 3) Model training:** Model used in this project is google's [Flan-T5](#) base with 250M parameters. The model was fine-tuned using selected hyper-parameters of batch size of 2, trained on 3 epochs, the Adafactor optimizer and 1.0 gradient clipping with model checkpoints to save and observe each epoch if necessary.
- 4) Evaluation:** The trained model was applied on the test set for the evaluation. The metrics used are ROUGE and BERT. ROUGE, measures lexical overlap between generated and existing summaries. BERT score evaluates semantic similarity using contextual embedding.

Pipeline Architecture

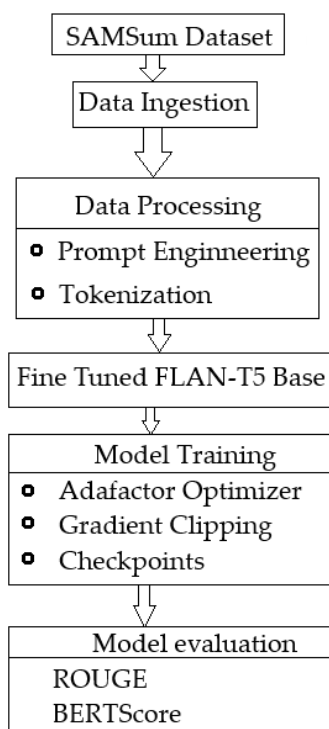


Figure 1: Pipeline Architecture

Hyperparameters

Hyperparameter	Value
Base Model	Google/flan-t5-base
Dataset	SAMSum
Epochs	3
Batch size	2
Optimizer	Adafactor
Max Input Length	384
Max Output Length	64
Gradient Clipping	1.0
Beam Size	4
Checkpointn Saving	Every Epoch

Table 1: Hyperparameter configuration used for model fine-tuning

Preprocessing

In this project the instruction based prompt engineering approach was applied to dataset along with tokenization. Each dialouge had an additional prompt on top that "Summarize the following conversation" followed by the conservation and a summay. This leverages the instruction following capabilities of the model via direct communication of the desired task.

After prompt construction the text was tokenized

Training and results

The Flan-T5 base model was fine-tuned on SAMSum dataset using the tuned parameters. The training process took 3 epochs with a batch size of 2 using the Adafactor optimizer and gradient clipping on 1.0 to ensure stable optimization. The model was trained using a method called teacher forcing, where the reference for the summarizers were provided during training learn the mapping between input and their corresponding results.

Evaluation metrics

1) ROUGE

Metrics	Score(%)
ROUGE-1	51.995625
ROUGE-2	27.048264
ROUGE-L	43.263412

Table 2: ROUGE results

the sequence and structure of the test data.

The fine tuned model achieved a ROUGE score of 51.99% indicating that more than half of the important works in the generated summary are similar to test data summary. The ROUGE 2 score of 27.09% indicate a moderate overlap of consecutive words in pair. It suggests that although the model has captured key data, some difference in the phrasing of the sentence remains. Lastly ROUGE-L, score 43.26 shows

good coherence and readability. It means the the generated summaries preserved

2) BERTScore

Metrics	Score
Precision	92.696261
Recall	92.221832
F1	92.440283

Table 3: BERTScore

The model achieved a high precision of 92.69, recall of 92.22 and a BERT F1 score of 92.44. These high value suggest that generated output is semantically very similar to the test output values that are human-written. Unlike Rouge, bert uses transformers embedding to evaluate contextual meaning, therefore the high BERTScore suggests the model sucessfully preserves the intent and meaning of original conservation.

Conclusion

The results positively demonstrates that fine-tuning FLAN-T5 base on SAMSum dataset produces efficient and desirable results and the model performannce based on how it conserves the data and key information is impressive. The high ROUGE score and high BERT score are proof of that. The key step I believe that improved the result in this project was prompt engineering approach as many small transformers are trained with "few shots learning", they provide significantly impressive results when an example is provided.

Despite the motivating preformance there are few limitations as well. The model sometimes omits less important words or produces summary with just slightly different wording from test summaries. The summary generated is mostly chunk of small sized sentences making it look robotic or primitive.

Future Work

Using larger T5 model like T5-large or T5-XL might improve the summarization but at a cost of high computing power and training time.

Fine tuning methods like LoRA or QLoRA could be investigated to reduce computational requirements while also maintaining performances.

Right now the model is good enough to be globally deployed. The code is structured that its resuable and it can easily fit in deployment pipeline. Here's an example [See the link in description](#).

References

1) FLAN-T5

Chung, H. W., Hou, L., Longpre, S., Zoph, B., Tay, Y., Fedus, W., et al. (2024). *Scaling Instruction-Fine tuned Language Models*. *Journal of Machine Learning Research*, 25(70), 1–53.

2)Attention Is all you need

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser & Polosukhin, I. (2017). *Attention Is All You Need*. In *Advances in Neural Information Processing Systems* (Vol. 30).